



OUR LADY OF FATIMA
UNIVERSITY

LESSON 1 INTRODUCTION TO

Cascading Style Sheet (css)





OBJECTIVES

At the end of the lesson, the students should be able to:

1. Define CSS (Cascading Style Sheets).
2. Explain the purpose of CSS in web development.
3. Identify the three ways of adding CSS to HTML (Inline, Internal, External).
4. Describe the basic CSS syntax (selector, property, value).
5. Recognize common CSS selectors (element, class, id).

What is CSS?

CSS stands for Cascading Style Sheets. It is a stylesheet language that is used to describe the visual presentation of a web page written in HTML (Hypertext Markup Language). HTML creates the structure of the page, while CSS adds styling to that structure.

Why the Term "Cascading"?

The term "cascading" refers to the way CSS prioritizes and applies styles when multiple rules target the same HTML element. This process considers specificity (how specific a rule is) and inheritance (styles passed from parent to child elements), ensuring the most relevant and recent rules take effect.

Why use CSS?

- **CSS Saves Time:** You can write CSS once and then reuse same sheet in multiple HTML pages.
- **Pages Load Faster:** If you are using CSS, you do not need to write HTML tag or attributes every time. Just write one CSS rule of a tag and apply it to all the occurrences of that tag.
- **Easy Maintenance:** To make a global change, simply change the style, and all elements in all the web pages will be updated automatically.
- **Superior Styles to HTML:** CSS has a much wider array of attributes than HTML, so you can get a far better look to your HTML page.
- **Multiple Device Compatibility:** For the same HTML document, different versions of a website can be presented for different screen widths
- **Global Web Standards:** Now most of the HTML attributes are being deprecated and it is being recommended to use CSS.

Key Features of CSS:

CSS offers a wide range of capabilities that make it indispensable for web development:

- Styles and layouts of web pages.
- Works alongside HTML and XML documents.
- Enables responsive design for different screen sizes.
- Supports interactive effects like hover states and animations.
- CSS is now modularized, with ongoing updates rather than version numbers.
- Allows reusability of the same rules across multiple HTML documents.

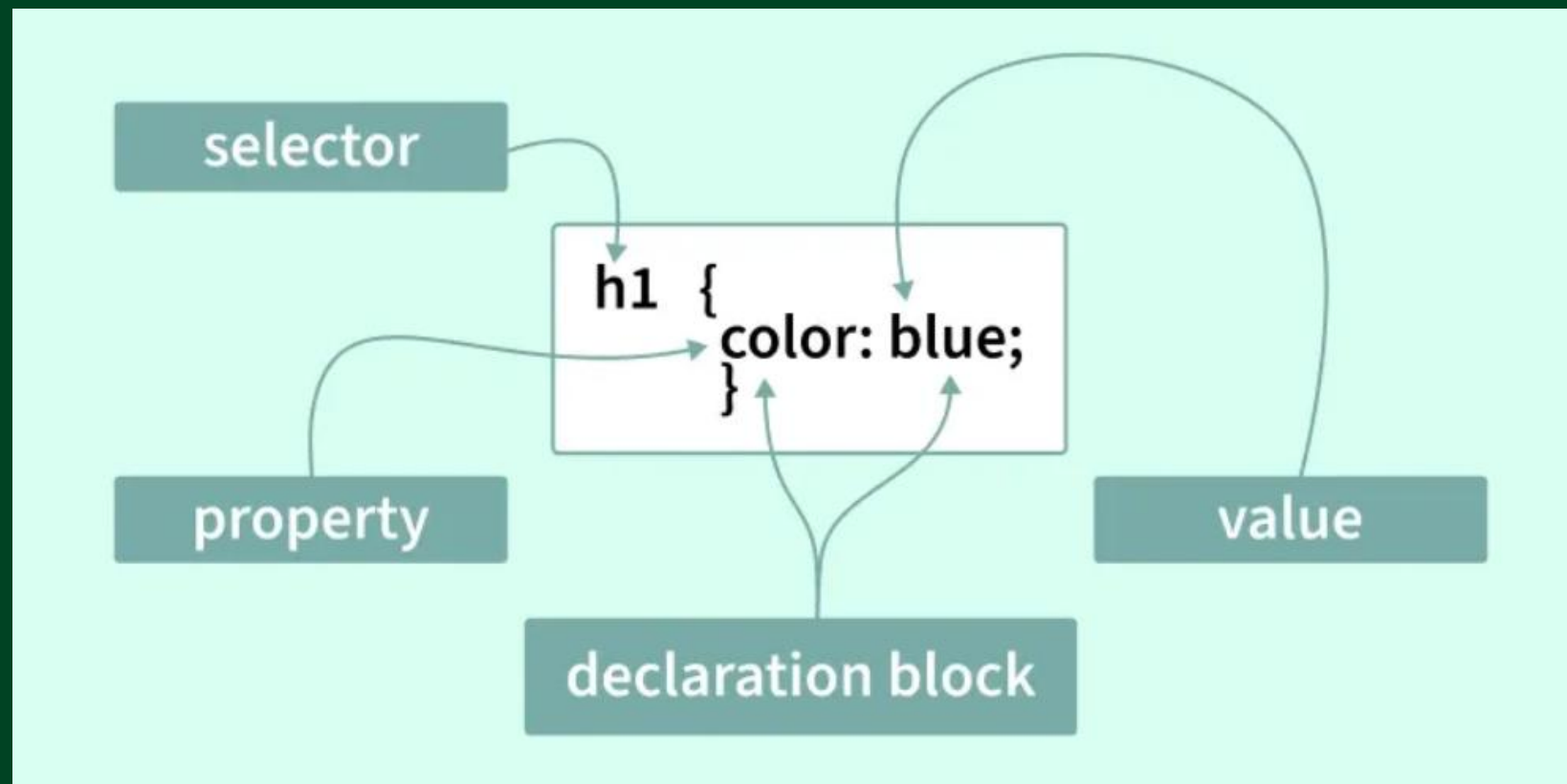
A Brief History of CSS



CSS was created by **Håkon Wium Lie** to enhance the visual aspects of websites. Initially, websites were mainly used by researchers and lacked visual appeal. As websites became more widespread, the need for better design grew.

CSS was released (in 1996), 3 years after HTML (in 1993). The main idea behind its use is that it allows the separation of content (HTML) from presentation (CSS). This makes websites easier to maintain and more flexible.

Syntax of CSS



1. **Selector:** Specifies the HTML element(s) to target (e.g., `h2`, `.class-name`, `#id-name`).
2. **Declaration Block:** Enclosed within curly braces `{ }` and contains one or more property-value pairs.
3. **Property:** Defines the style attribute to modify (e.g., `color`, `background-color`).
4. **Value:** Sets the specific style for the property (e.g., `blue`, `5px`).
5. **Semicolon (;):** A critical component, required at the end of each property-value pair to separate multiple declarations within a block.



Ways to Add CSS

There are three different ways to add CSS to an HTML page, which are:

1. **Inline CSS**
2. **Internal CSS**
3. **External CSS**



Inline CSS

Inline CSS is used to add custom properties to specific elements. The added style will only reflect on that particular element only.

Implementation To use inline CSS, insert the style attribute within the HTML element's opening tag.

```
<h1 style="color: purple;"> I'm Arjay </h1>
```

```
<h2> Welcome to Web Development 1 </h2>
```

Internal CSS

Internal CSS is used to apply custom style to multiple elements on the same page. The style can be used throughout the HTML page.

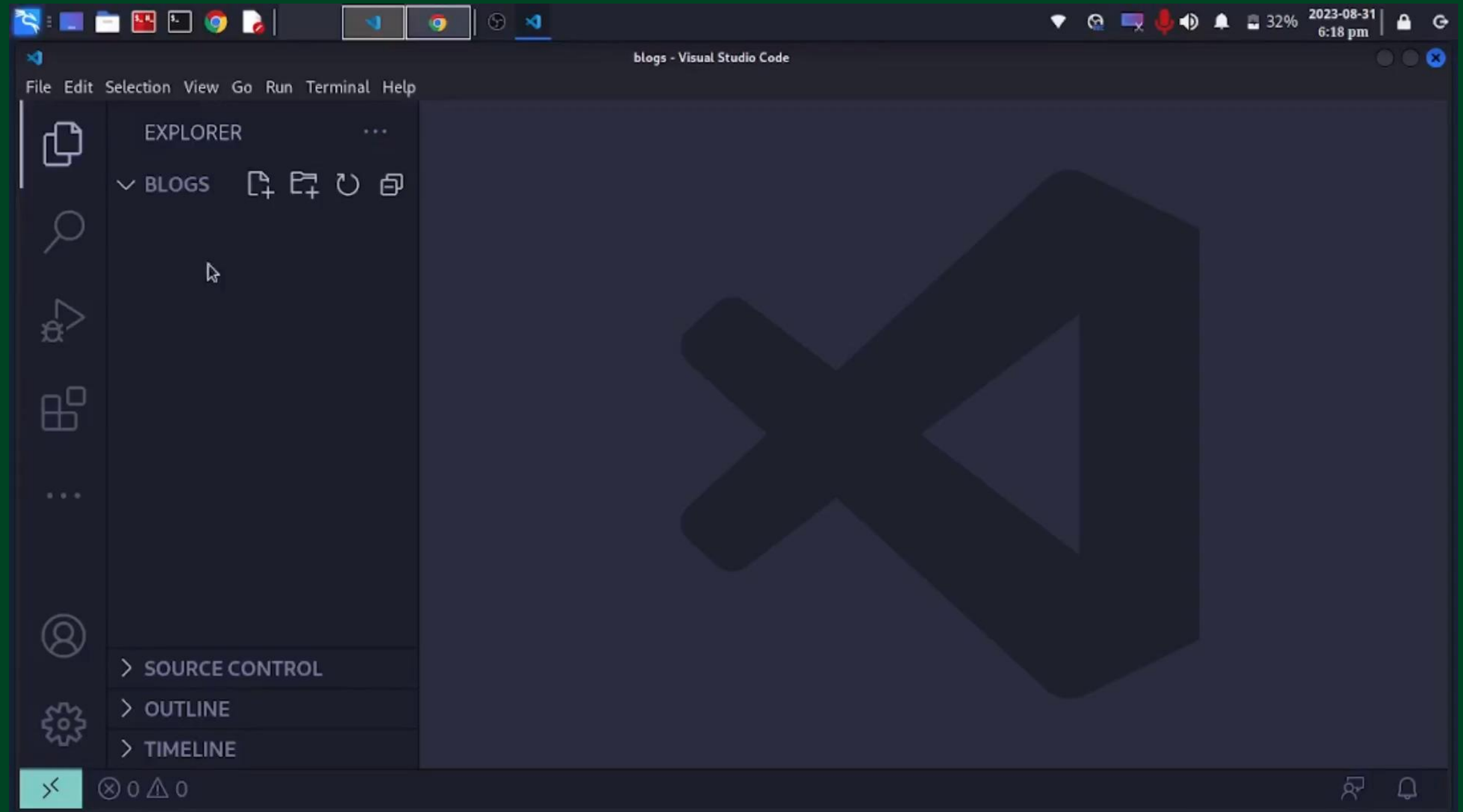
Implementation Internal CSS is defined in a style block, which will be inside the head section.

```
<head>
  <style>
    p {
      color: red;
    }
  </style>
</head>

<body>
  <h1> Welcome to Web Development 1 </h1>
  <p> I'm Arjay, from CCS Department </p>
  <p> I love Information Technology </p>
</body>
```

External CSS

External CSS works similarly to internal CSS but with a twist. Instead of adding the styles within the HTML file, we create a separate file with a **.css** extension. This file will hold all the styling details. Then, we link this file to the HTML page, giving it the instructions on how to look.



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```
<html lang="en">
```

```
<head>
```

```
<title> External CSS </title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```


Comments

Comments in CSS play a crucial role in improving code readability and maintainability, serving as valuable documentation for developers who review or modify the code in the future.

Comments are ignored by the browser and don't affect the styling or layout.

CSS Comments are enclosed between `/*` and `*/`.

There are two types of comments in CSS:

1. **Single-line comment** – Single-line comments are contained within one line. They are useful for short annotations.
2. **Multi-line comments** – Multi-line comments span across multiple lines, making them ideal for detailed explanations or temporarily disabling blocks of code.



Selectors

CSS selectors allow us to choose specific elements and apply styles to them. Suppose we want to add a custom style to only a specific tag(s). There, we can make use of CSS selectors.

There are different types of CSS selectors, which are as follows:

1. Universal Selector
2. Element Selector
3. Id Selector
4. Class Selector
5. Group Selector

Universal Selector

Universal selector represented by ***** targets all the HTML elements on the page.

```
* {  
    property: value;  
}
```

```
<html>  
<head>  
    <style>  
        * {  
            color: purple;  
            text-align: center;  
        }  
    </style>  
</head>  
<body>  
    <p> Welcome to Web Development 1 </p>  
    <h1> I love Web Programming </h1>  
</body>  
</html>
```

Element Selector (Type Selector)

The element selector selects the target element based on the specific type. Suppose you want to underline all the `<p>` tags; in this case, the element selector will be the best choice.

```
p {  
  property: value;  
}
```

```
<head>  
  <style>  
    p {  
      text-decoration: underline;  
    }  
  </style>  
</head>  
<body>  
  <h1> Web Development 1</h1>  
  <h2> we offer: </h2>  
  <p> Python Tutorials – 100 Days of Code </p>  
  <p> Ultimate JavaScript Course </p>  
  <p> React JS Tutorials For Beginners </p>  
</body>
```

ID Selector

The ID selector targets the elements based on the specific ID. It is written with the hash # character followed by the ID name in the style sheet.

```
#ID {  
    property: value;  
}
```

```
<head>  
    <style>  
        #title {  
            text-align: center;  
            color: red;  
        }  
    </style>  
</head>  
<body>  
    <h1 id="title"> Web Development 1</h1>  
    <p> I love Web Programming </p>  
</body>
```


Class Selector

The class selector does the same job as the id selector, a class selector helps group various types of elements. Suppose, we want to give a custom style to a specific group of elements. In this case, the class selector is the best option. It is written with the period . character followed by the class name in the style sheet.

```
.class {  
    property: value;  
}
```

```
<head>  
    <style>  
        .red {  
            color: red;  
        }  
    </style>  
</head>  
<body>  
    <p>This is simple p tag</p>  
    <p class="red">This p tag has class red</p>  
    <p>This is simple p tag</p>  
    <p class="red">This p tag has class red</p> </body>
```

Group Selector

The group selector is used to minimize the code. Commas , are used to separate each selector in a grouping. This reduces the number of lines of code. The code also looks clean.

```
div, p, a {  
    property: value;  
}
```

```
<head>  
  <style>  
    h1 {  
      color: red;  
    }  
    p, a {  
      color: purple;  
    }  
  </style>  
</head>  
  
<body>  
  <h1>Web Development 1</h1>  
  <p>This is the p tag</p>  
  <a href="#">This is the anchor (a) tag</a>  
</body>
```



Selector Summary:

- Universal Selector (*): Target the entire page.
- Element Selector: Target a specific element.
- ID Selector (#): Target element with a specific ID.
- Class Selector (.): Target element(s) with the same class.
- Group Selector: Group elements and target them.

With selectors, CSS gives you the precision to style your page exactly how you want—whether it's one element, a group of them, or even just the first letter of a paragraph.



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TAKE AWAYS

“CSS is not just about making a website beautiful – it is about making it usable, readable, and professional.”